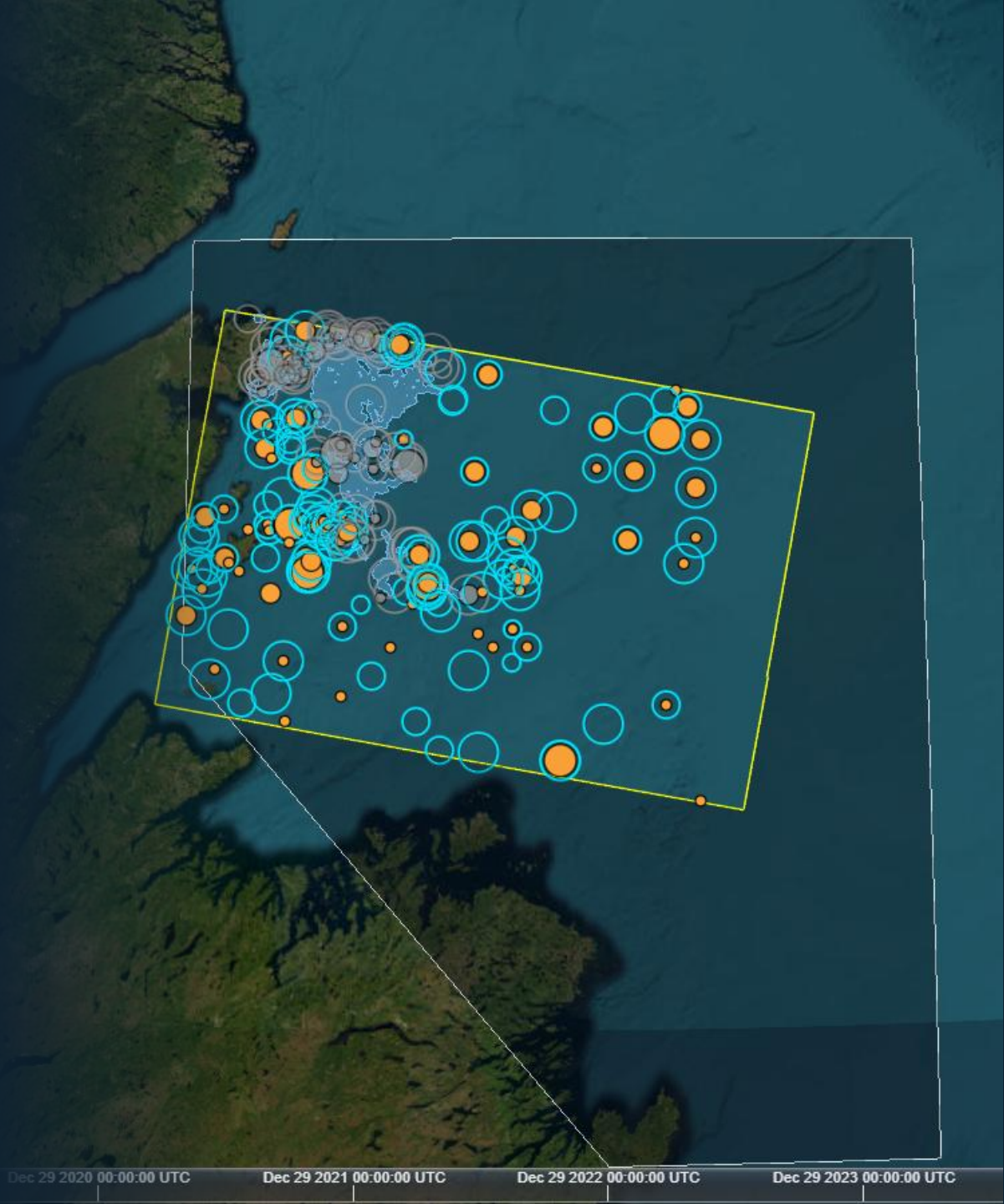
BRIEF · ICEBERG DETECTION

# CA-CFAR vs. THOR

A classical radar detector and a SAR foundation model, tested on the same scenes against the same Ice Patrol labels.

Natali Kovalski · October 2026



THE CONTENDERS

# Two ways to find a bright berg

## CA-CFAR

- **No training:** each pixel vs. its local background, on HV
- **Threshold** from a false-alarm rate and the scene's speckle
- **Filters:** size, ice strips up to 300 m, HH co-pol check
- **CPU:** a full 8-PFA sweep takes about 6 min per scene

## THOR + linear head

- **Frozen encoder:** THOR Tiny (FM4CS), pretrained on Sentinel-1
- **769-parameter head** trained on IIP labels, 2019–2020
- **Input:** HH/HV sigma0 in dB, 80 m patches
- **GPU:** 8–15 min per scene on a laptop RTX 3070

TEST SETUP

# Everything identical except the detector

## Same frames

Four April 2019 passes with same-pass IIP analyst labels: 512 bergs. None was used in training.

## Same masks

Land + 500 m, pack ice + 1 km. Detections within 5 km of pack ice are dropped for both.

## Same matching

One-to-one within 500 m, with IIP positions moved to their 0.1′ cell centres.

## Same budget

Precision needs AIS, so recall is compared at a matched number of detections per frame.

HEADLINE RESULT

# +10 pts

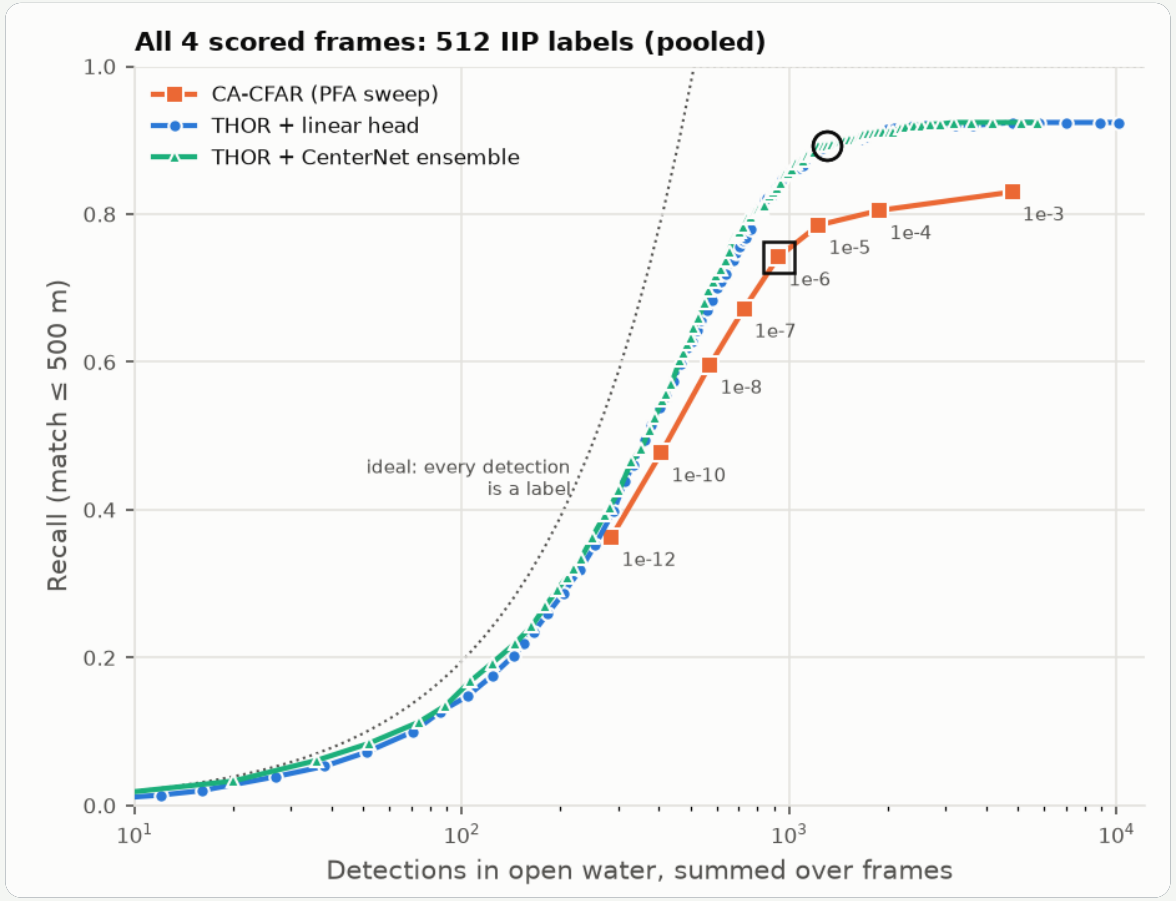
THOR finds 84% of the labelled bergs where the CA-CFAR finds 74%, at the same number of detections: **431 vs. 380 of 512**.

4 held-out frames · 926 open-water detections each · THOR leads on every frame

RECALL VS. DETECTIONS

# THOR climbs, CFAR saturates

- **Same recall** as the CFAR with ~28% fewer detections
- **CFAR at its loosest** (PFA $10^{-3}$): 0.83 with 4,798 detections
- **THOR** reaches that with ~900, and tops out at 0.92
- **Its top ~100 peaks** are almost all labelled bergs




Pooled, 512 IIP labels. Square: CFAR at PFA $10^{-6}$. Circle: THOR at score 0.35.

PER FRAME

# Ahead on every frame, by 0 to 15 points

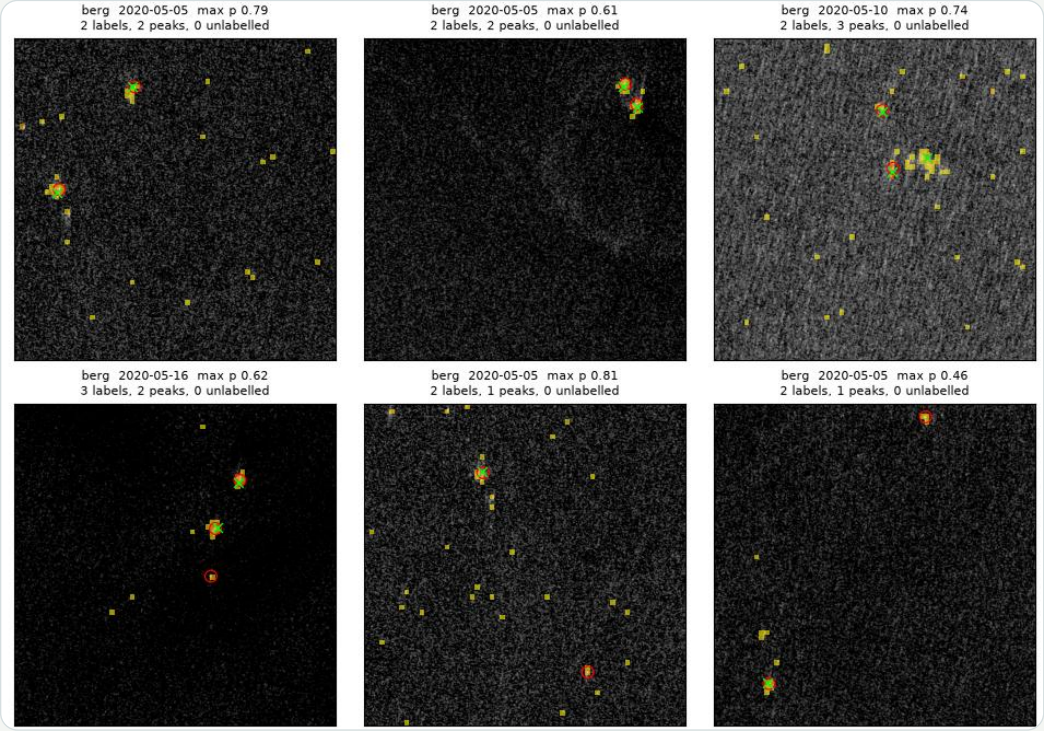
| Frame | IIP labels | Detections | CA-CFAR | THOR |
|---|---|---|---|---|
| 2019-04-29 | 267 | 476 | 0.74 | 0.89 |
| 2019-04-23 | 150 | 278 | 0.79 | 0.82 |
| 2019-04-24 | 49 | 44 | 0.76 | 0.82 |
| 2019-04-17 | 46 | 128 | 0.61 | 0.67 |
| Pooled | 512 | 926 | 0.74 | 0.84 |

04-17 is low for both: 7 of its 46 bergs lie within 5 km of pack ice, where both detectors are filtered out. The first frame, 04-29, overstated the gap.

WHERE THE GAIN IS

# Large and medium bergs, not small ones

| Bergs (pooled) | CFAR | THOR |
|---|---|---|
| Small, 15–60 m (73) | 0.71 | 0.71 |
| Medium, 61–120 m (425) | 0.76 | 0.87 |
| Large, > 120 m (14) | 4 | 11 |
| > 20 km from ice (275) | 0.80 | 0.93 |
| Incidence 40–47° (398) | 0.73 | 0.83 |




THOR heatmaps on 2020 validation chips: IIP labels (red circles), THOR peaks (green ×). Large bergs: the CFAR sees them, then its 300 m strip rule drops them.

LIMITS

# THOR doesn't replace the pipeline

01

## Pack ice

Alone, it fires on pack ice: 34–82% of its peaks on icy frames fall inside the CFAR's ice mask.

02

## Small bergs

A tie at 0.71. An 80 m patch is coarse for a 20 m target; 4×4 patches are the next test.

03

## Precision

Unknown for both without AIS. At the matched count, CFAR has 546 and THOR 495 unlabelled detections.

VERDICT

# THOR on top of the CFAR pipeline

## What holds

- **+10 points recall** at an equal number of detections
- **Large bergs:** 11 of 14 found, CFAR 4
- **Needs the pipeline:** its pack-ice mask, and a GPU
- **Caveat:** one month of one year; THOR learned the analysts' labels

## Next

- **AIS for the test frames** to measure precision for both
- **Shape-based strip test** so the CFAR keeps large bergs
- **THOR at 4×4 patches** for small bergs
- **More years** and near-range frames

THANK YOU

# Questions?

Natali Kovalski · github.com/natali-kovalski/iceberg-alley-sar

Live viewer with both detectors: natali-kovalski.github.io/iceberg-alley-sar. Write-up: docs/thor-vs-cfar.md. Contains modified Copernicus Sentinel data (2019–2026). Labels: International Ice Patrol (NSIDC G00807). THOR: FM4CS (Apache 2.0); Forgaard et al., IGARSS 2026.